CNN-based Voice Activity Detection
Diagnostic Report (Real Data Training)

1 Objective

The goal of this experiment is to diagnose the failure of a CNN-based Voice Activity Detection (VAD) model trained on synthetic data when applied to real audio, and to evaluate whether training on real speech data resolves the issue.

2 Background

Previous experiments showed that a CNN trained on synthetic data completely fails on real audio:

- False Alarm Rate (FAR) = 1.0
- All frames predicted as speech

Feature analysis revealed a significant distribution mismatch between synthetic and real audio inputs.

3 Feature Distribution Analysis

3.1 Synthetic Data

Synthetic data exhibits a compact and structured distribution.

3.2 Real Audio

Real audio shows significantly higher variance and a very different distribution.

3.3 Comparison

Key observations:

- Real data has much larger dynamic range
- Mean shift is significant
- Distributions barely overlap

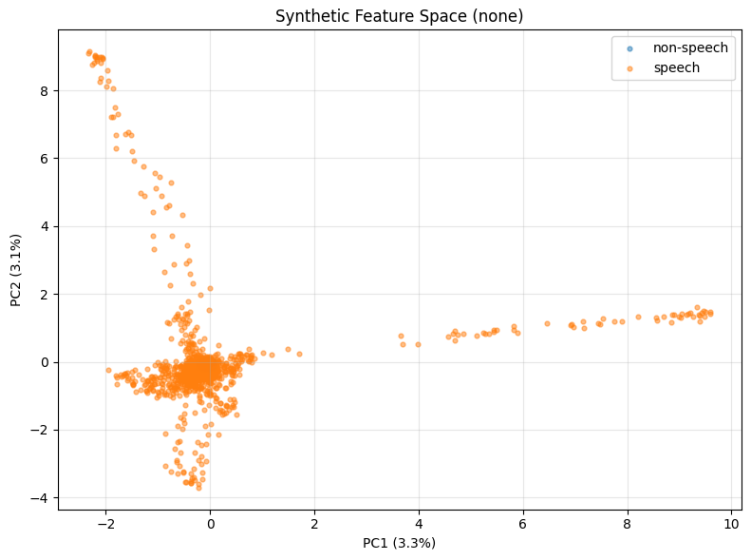

Figure 1: Synthetic feature space (PCA)

4 Training on Real Data

The CNN model was retrained using real speech data (LibriSpeech).

- Best validation F1: 0.9184

This indicates that the model successfully learns meaningful representations from real data.

5 Evaluation on Real Audio

5.1 Default Threshold (0.5)

- F1 Score: 0.0500
- FAR: 0.0000
- MISS: 0.9744

The model predicts almost all frames as non-speech.

5.2 Probability Distribution

Statistics:

- Mean ≈ 0.09
- Most predictions in $[0.02, 0.15]$

Observation: The model outputs are systematically biased toward low probabilities.

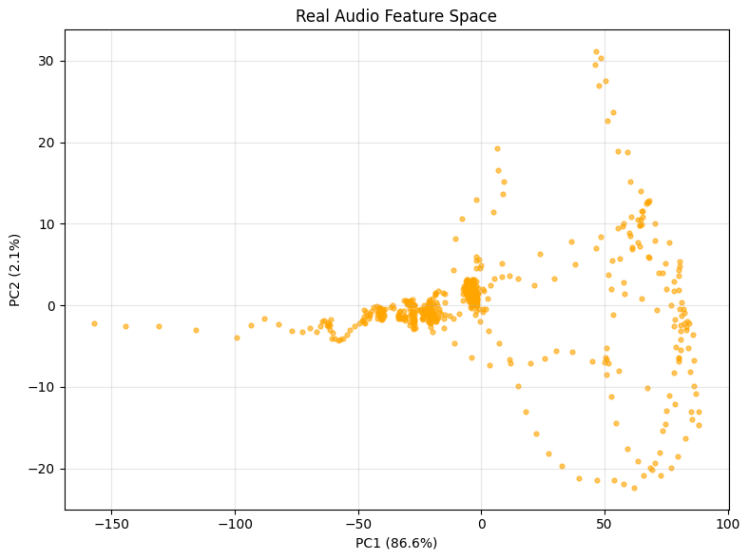

Figure 2: Real audio feature space (PCA)

6 Threshold Sensitivity

After tuning the threshold:

- Threshold = 0.1
- F1 Score = 0.6169
- FAR = 0.1046
- MISS = 0.4167

This shows that:

- The model has learned useful discriminative features
- The issue lies in decision threshold rather than representation

7 Qualitative Results

- Predictions roughly align with speech segments
- Some fragmentation and missed detections remain

8 Conclusion

- Training on synthetic data fails due to severe domain mismatch
- Training on real data resolves the collapse issue
- However, the model outputs poorly calibrated probabilities

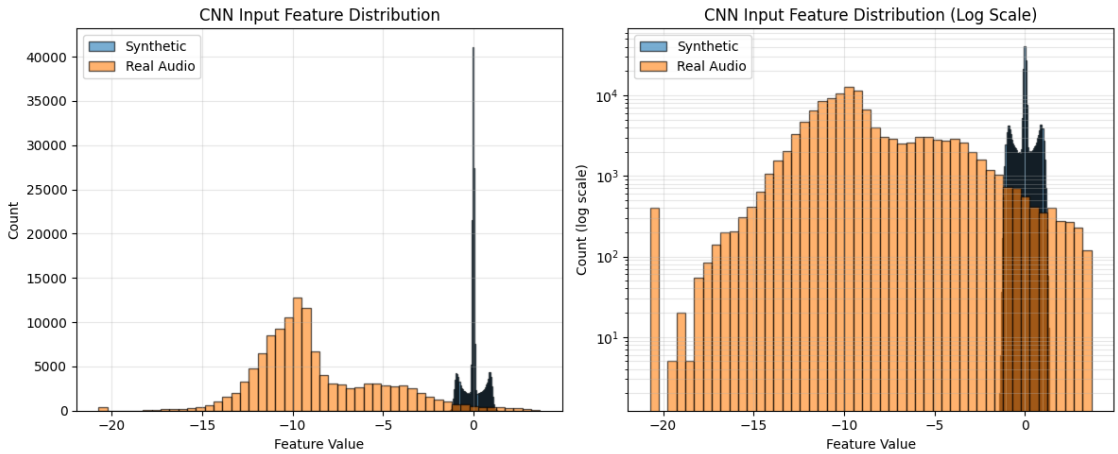

Figure 3: Feature distribution comparison (synthetic vs real)

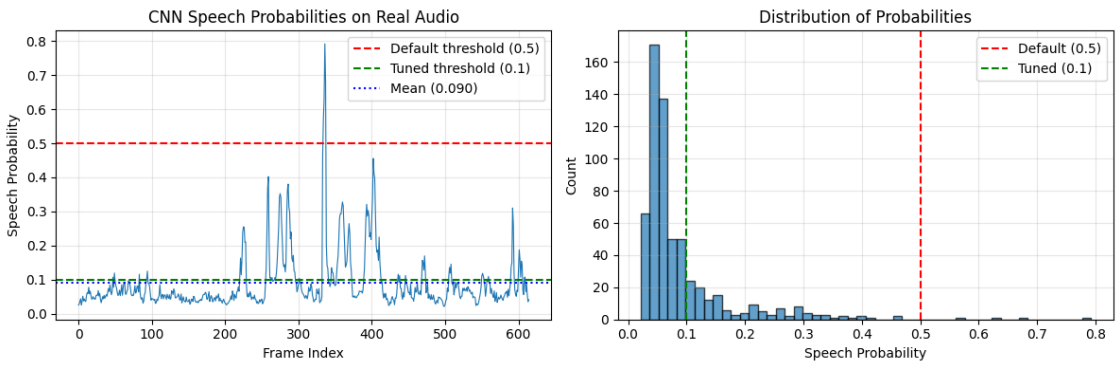

Figure 4: CNN output probability distribution

- Default threshold (0.5) is not suitable
- Threshold tuning significantly improves performance

Final insight:

The CNN model trained on real data is capable of distinguishing speech from non-speech, but requires proper calibration or threshold selection for effective deployment.

9 Future Work

- Apply probability calibration (Platt scaling / temperature scaling)
- Improve class balance during training
- Incorporate more diverse noise (e.g., MUSAN)
- Explore loss reweighting strategies

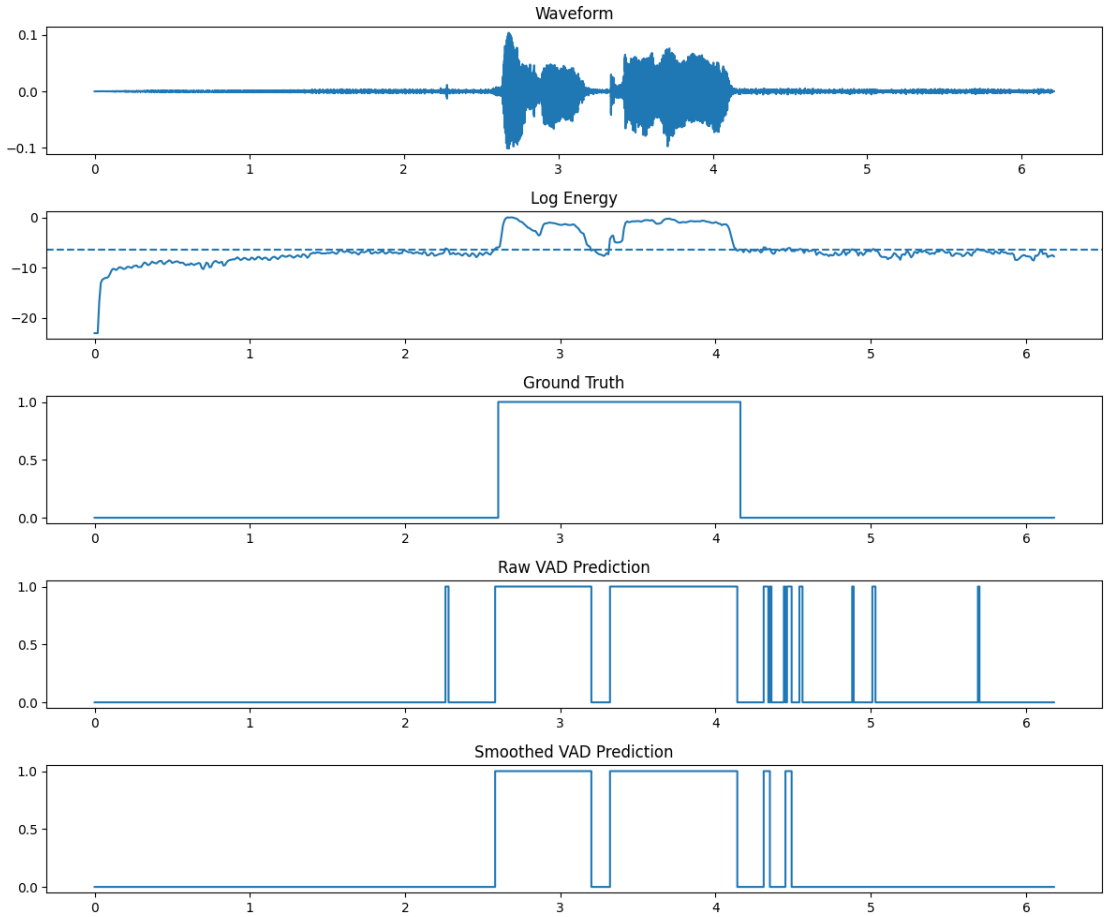

Figure 5: VAD prediction vs ground truth